

Quality in Fume Filtration, [IP System USA](#)). It should be noted that VOCs require certain filters, "While these systems are effective for epoxy fumes, hazards from VOC (Volatile Organic Compounds) such as perfumes may require you come up with a VOC fume extraction strategy such as using quality carbon filters." (IP Systems Superior Quality in Fume Filtration, [IP System USA](#)). While 3D pens seem to be harmless compared to resin, they also emit toxic fumes. When using ABS a plastic smell can often be admitted, "Some users notice a slight "plasticity" smell when using ABS." (3Doodler, [3Doodler](#)). The Doodler also emits a smell of PLA, which can smell like syrup. "Many users prefer the smell of PLA, which, being corn based, can smell like popcorn or syrup." (3Doodler) Even though the scent seems pleasant, the official site recommends using protection, in other words a ventilation system. "No matter what material you're using, and whether you're using a 3Doodler or a traditional 3D printer, we recommend working in a ventilated area." (3Doodler, [3Doodler](#)). However, you can always change to the 100% non-toxic version, which releases no fumes. "The 3Doodler Start and 3D Build & Play Eco-Plastics are 100% non-toxic and biodegradable in household composts, so there are absolutely no fumes" (3Doodler, [3Doodler](#)). Glass can also be harmful to inhale, especially its dust. Glass is made from a chemical known as "silica dioxide," which is not harmless. "Glass is manufactured using sand, otherwise known by its chemical name "silica dioxide", however due to the heating and rapid cooling techniques used in the creation of glass, both glass and glass dust are examples of amorphous silica and therefore are relatively harmless." (Krysteline, [Krysteline](#)). However crystalline silica can cause multiple respiratory issues. "Respirable Crystalline Silica (RCS) is released as a very fine dust. The inhalation of which is hazardous and well documented to cause silicosis and myriad respiratory disorders. Crystalline silica when inhaled is classified as carcinogenic to any animal or human by The World Health Organization." (Krysteline, [Krysteline](#)).

5) Where is the problem happening?

The problem is mainly occurring in Mrs. Hudson's and Mrs. Poole's classrooms. However these pollutants spill into the art hallway, but don't pour into the main hallways.

6) When does the problem occur?

The problem occurs mainly during the weekdays. (Mrs. Poole) Most of the time when they are doing projects with resin, 3d pens/ printing, and glass. The 3d pens release a harmful gas into the air when they are in use, as they are burning the plastic (PLA or ABS). While PLA is less harmful, it still produces an odor and gas. The resin has been described to have a "very bad smell". Finally, when using glass, students often have to cut it with a dremel, which leaves micro glass particles into the air around them. (ScienceDirect, [source](#)).

7) Why does the problem happen?

This problem happens whenever the students use 3d pens or resin with Mrs. Poole, or when the

	students are cutting glass or kilning with Mrs. Hudson. While doing these things aren't directly harmful in a small amount, when a whole class is doing these things at once, the air becomes very polluted. When they are cutting glass, even though they may not see it, there are a lot of micro particles released into the air. "Nowadays nano/micro-plastic pollution gets enormous attention between scientists and researchers. Plastic and glass pollution are considered the worst in current situation. As glass is endlessly recyclable and never loses its quality and purity no matter how many times it is recycled" (Science direct, source). 8) How often does the problem happen? This problem happens whenever Resin is in use "Volatile organic compounds (VOCs) are emitted as gasses from certain solids or liquids. VOCs include a variety of chemicals, some of which may have short- and long-term adverse health effects"(EPA) (source) , glass is being grinded "Symptoms of silicosis usually appear after many years of exposure. In early stages, symptoms are mild and include cough, sputum and progressive shortness of breath. As the scarring continues to worsen, the first real signs of a problem may be an abnormal chest X-ray and a slowly developing cough".(American Lung Association)(source), 3d pens are in use "The filaments with or without additives appear benign except for copper, which causes higher changes in stress, cell death, and metabolic perturbations".(ACS Publications) (source). , or when 3d models are being sanded down. In order to earn a score of 5, you will need to cite at least five sources that are from the last ten years and that come from 3 or more of the following credible categories: peer reviewed journals, professional organizations, government agencies, local and national newspapers. Use in-line citations to explicitly state which information is from which source. Do not just list all of the citations at the end of the submission or in footnotes. Grayson Michael Camden
Stakeholder Concerns and Connections (A3, A4)	Directly indicate at least five primary stakeholders or stakeholder groups. These are the people most directly impacted by the problem. Highlight the concerns of your primary stakeholders and use that to justify why there is a need to address the problem. Consider the breadth of the problem and how it might have social and environmental impacts on secondary stakeholders, which are people who may be impacted by a solution to the problem and are still part of the system. The primary stakeholders are Mrs. Poole and Mrs. Hudson. Mrs. Poole needs an air filter for the gasses produced from 3d printing and the odor released from resin, while Mrs. Hudson needs the glass particulates and harmful air from kilning filtered out. Mrs. Poole has stated the 3d pens "produce a foul odor". While the PLA scent is much more pleasant while it's being burned, she would like both the PLA and ABS scents minimized. The resin produces a foul odor too while it's in use, especially when multiple people are working with it at once. Mrs. Hudson said the bad scent from kilning and the dust from the glass are the main problems wanting to be solved.

